DealLens: Human-AI Decision Support for In-Progress Sales-Engineering Risk Sensemaking

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Abstract

Sales Engineers translate technical capabilities into customer-specific solutions and shape whether complex B2B opportunities progress, yet they must interpret active deal risk from fragmented CRM outcomes and conversation signals. We present DealLens, a human-AI decision-support system that links Salesforce opportunities with Gong interactions [1,2], estimates cumulative Loss risk from Call 4 onward, and makes evidence available for SEs and managers to inspect before acting. The predictive pipeline linked 1,868 labeled opportunities and retained 195 with more than three calls. An interpretable Logistic Regression model detected 82.5% of eventual deal losses at Call 4, reaching 93.2% at Call 8. Two stakeholder sessions informed design goals for integrated evidence, early sensemaking, actionable diagnostics, and human control. The complete four-module web interface includes progressive risk monitoring, signal diagnostics, and a conversational workspace for querying, editing, and contesting guidance. Predictive performance has been evaluated; the effects of these interactions on professional judgment await a domain-expert study.

CCS Concepts

Human-centered computing → Interactive systems and tools; HCI design and evaluation methods; Computing methodologies → Artificial intelligence.

Keywords

human-AI interaction, human-AI decision-making, explainable AI, mixed-initiative interaction, human agency, evidence-based decision support

1 Introduction

Sales Engineers (SEs) occupy a consequential position in complex B2B sales: they translate technical capabilities into customer-specific solutions, address technical objections, and often shape whether an opportunity progresses. The evidence needed to judge that progress is split across complementary systems. Salesforce records opportunity ownership, stage, pipeline state, and final win/loss outcome [1]. Gong captures customer interactions across calls, email, and web meetings and applies AI to surface conversation signals and revenue guidance [2]. Gong therefore shows how an engagement unfolds, whereas Salesforce records what happens to the opportunity; reviewed separately, they do not readily connect technical behavior to deal outcome or support action before closure.

This separation creates a human-AI interaction problem, not only a prediction problem. Useful assistance must surface risk early enough to change behavior, explain evidence in terms meaningful to SE practice without overwhelming users with an unfamiliar feature space, and preserve professional judgment when system suggestions conflict with deal context. These safeguards matter because a false high-risk alert can waste scarce specialist attention, while an overly confident recommendation can make correlation look like causal advice.

To address this gap, we developed DealLens, an evidence-grounded human-AI decision-support system that joins Salesforce outcomes with Gong interaction records, produces cumulative Loss-risk estimates from Call 4 onward, and presents trajectories, signal diagnostics, and a conversational next-action workspace within a shared SE/manager workflow (Figure 1). This paper contributes: **(1) Progressive Risk Prediction:** An operational ML pipeline integrating 1,868 opportunities and 2,029 call records, capturing evolving deal-risk trajectories with 82.5% Loss recall at Call 4 and 93.2% at Call 8.**(2) Human-Centered Decision Support Design**: Four stakeholder-derived design goals translated into an evidence-grounded architecture that distinguishes source evidence, model inference, AI-generated guidance, and human judgment. **(3) Interactive Human-AI Decision Support**: An implemented four-module interface enabling progressive risk inspection, evidence-based dialogue, and revision of AI recommendations, with a planned domain-expert ablation study examining appropriate reliance and decision ownership..

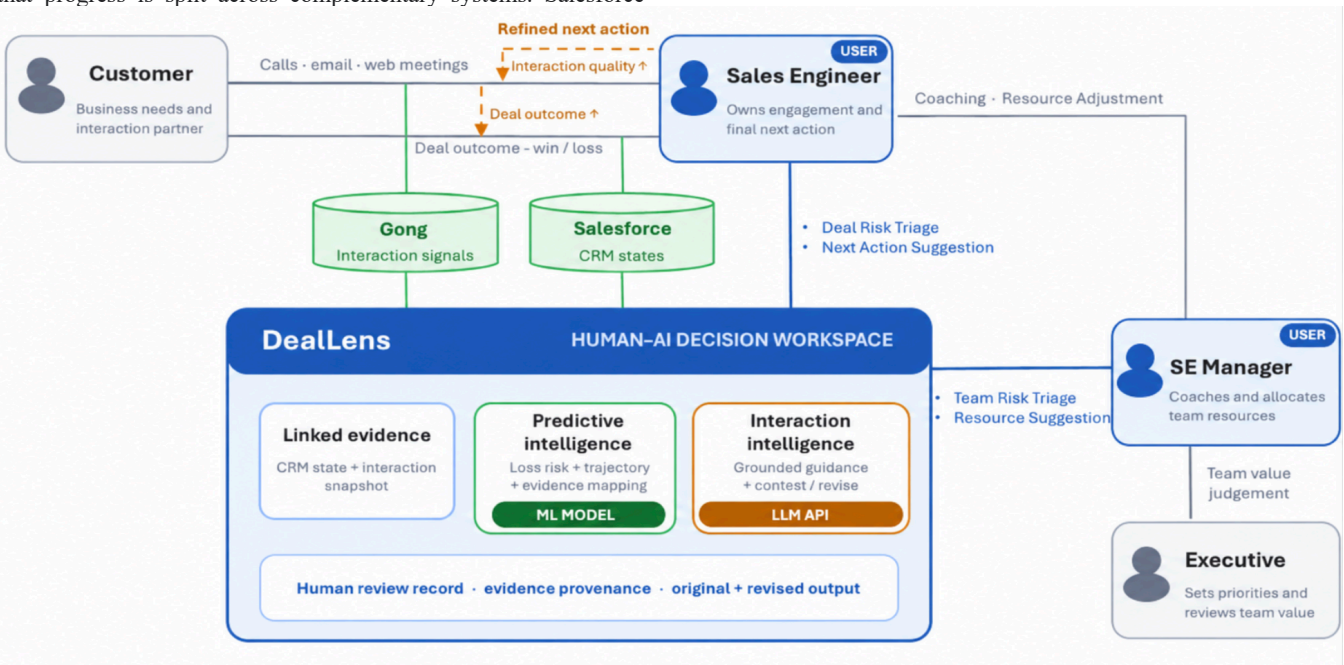


Figure 1: DealLens system architecture connecting customer interactions and enterprise records to linked evidence, predictive risk, grounded guidance, and human-owned Sales Engineer and manager decisions.

2 Related Work

Enterprise sales analytics and evidence-grounded risk. Salesforce and Gong provide complementary records of opportunity states and customer interactions [1,2]. Sales intelligence systems extract conversational signals and support coaching, as illustrated by SmartSales [3], while pipeline analytics predicts outcomes from structured data [4]. However, these approaches offer limited support for tracking how evidence accumulates, why risk changes, or how professionals should respond to recommendations. Bhattacharya et al. [5] introduced directive data-centric explanations combining risk predictions with interactive what-if exploration to help healthcare professionals understand diabetes risk factors and potential actions. While their work emphasizes interpreting risk factors and hypothetical changes, DealLens examines how evidence from successive customer interactions informs evolving risk assessments and professional decisions. Rather than treating conversational features as causal interventions, DealLens connects cumulative risk trajectories with inspectable evidence and revisable guidance. Research on uncertainty further highlights how probabilistic advice influences decision-making [6].

Explanation, interaction, and appropriate reliance. Human-AI guidelines emphasize system transparency, correction, and user control [7], while question-driven XAI addresses practitioners' information needs [8]. Research on explanatory debugging and user-contributed rules demonstrates the value of interrogating and correcting intelligent systems [9,10]. DealLens implements these principles in a conversational decision-support UI that lets practitioners inspect and revise guidance without modifying the prediction model. Studies of cognitive forcing [11] and appropriate reliance [12] caution that explanations and trust do not guarantee better judgment. Explanations may fail to improve complementary performance [13], while model-behavior descriptions and verification tools can help identify unreliable advice [14,15]. DealLens therefore distinguishes source evidence, model inference, generated guidance, and human overrides.

Professional collaboration and accountability. Organizational research characterizes AI assistance as a complement to situated human expertise [16] and examines the opportunities and tensions of AI as a collaborator [17]. Recent studies of human-AI collaboration underline the importance of professional involvement and accountability [18] and show how continuing evidence support can be more useful than a recommendation alone in complex professional decisions [19]. At the same time, algorithmic management scholarship warns that behavioral metrics may become decontextualized instruments of control and workplace surveillance [20,21]. DealLens consequently frames call metrics as deal-level diagnostic cues, not individual SE performance ratings, and reserves coaching and staffing decisions for professionals. Finally, Vipera demonstrates a visual-plus-LLM approach to inspectable AI analysis and feature-ablation evaluation [22]; DealLens implements the conversational UI and proposes a related progressive evaluation; the summative user study remains future work.

3 Formative Study and Design Goals

We conducted one in-person and one remote stakeholder session with an industry partner. Attendees represented four roles: an SE team member, an SE manager, a data-team manager, and executives (CEO and CTO). Discussions examined how the organization understands SE contribution, why Salesforce and Gong evidence is difficult to use together, and what assistance could help during an active deal. These sessions originated as UCI capstone requirements discussions; we therefore report consolidated needs rather than claiming a formal interview protocol, recordings, or thematic saturation. Table 1 summarizes the four findings and resulting design goals.

Table 1: Four formative findings and the corresponding design goals that translate fragmented enterprise evidence, late risk awareness, coarse scores, and missing context into concrete system requirements.

Finding	Design goal and response
F1: Teams need credible evidence of SE contribution, but outcome and interaction data are fragmented.	DG1 - Integrate outcome and behavior evidence: link Salesforce deal context to Gong interaction signals.
F2: Post-hoc explanation arrives too late to change the engagement.	DG2 - Support early, longitudinal sensemaking: begin cumulative risk inspection once sufficient call history exists.
F3: A coarse risk score does not reveal which interaction behaviors need coaching.	DG3 - Make risk inspectable and actionable: show salient signals, reference ranges, source evidence, and action cues.
F4: SEs hold missing context; managers must allocate scarce attention.	DG4 - Preserve human control: enable follow-up questions, notes, editable guidance, and manager triage.



Figure 2: Technical prediction pipeline implementing DG1 and DG2: Salesforce outcomes are joined with Gong interaction records, filtered to stable histories, transformed into cumulative call snapshots, and scored with an inspectable logistic-regression model.

4 System Design

4.1 ML Prediction Layer

This layer directly operationalizes DG1 and DG2. In response to DG1, it integrates Salesforce outcome context with Gong behavioral evidence at the deal level. In response to DG2, it reconstructs cumulative Call-4-onward snapshots so practitioners can inspect how risk changes while customer action remains possible (Figure 2).

Data linking

The source dataset contained 1,868 labeled opportunities: 753 Win and 1,115 Loss. We matched each Salesforce opportunity to Gong by Deal ID, then linked its conversation, call, and email-level records through conversation keys. Requiring at least four completed calls retained 195 opportunities (81 Win, 114 Loss), providing enough interaction history for stable behavioral signals.

Model selection

We compared Logistic Regression (LR), Random Forest, and Decision Tree classifiers across three feature sets: all available signals, call-derived signals only, and email-derived signals only, with Loss as the target class. Call-derived signals achieved the strongest predictive performance, highlighting customer-call interactions as particularly informative indicators of evolving deal outcomes. LR was selected for its stable performance on the small cohort and interpretable feature contributions. Using call-derived signals, LR achieved 0.857 precision, 0.783 recall, 0.818 F1, and 0.861 ROC-AUC. We further rank signals by their coefficient-based associations with deal outcomes; prominent features, including question rate, call completion, longest monologue, and talk ratio, inform the Interaction Signal Diagnostics module. These findings motivate closer examination of conversational behavior throughout

the sales process, while treating observed associations as diagnostic evidence rather than causal effects.

Cumulative snapshots for active deals

Training only on completed deals would overlook how risk signals evolve during the sales process. We therefore construct cumulative snapshots at sales call milestones to capture stage-dependent nuances in interaction patterns and their associations with eventual outcomes. For a deal with six calls, three training samples represent Calls 1–4, 1–5, and 1–6, each labeled with the final win/loss outcome. This allows the model to learn how accumulating conversational evidence relates to deal risk at different stages. At inference time, DealLens updates the risk estimate after each new call using all evidence available to that point, revealing emerging patterns and shifts in risk trajectories rather than isolated predictions. We begin at Call 4 to ensure sufficient behavioral evidence. The model uses nine interaction signals and a call-progress indicator across 2,029 call records. Grouped cross-validation prevents deal-level leakage, while opportunity-balanced weights prevent longer deals from dominating training. This design enables longitudinal risk monitoring and timely professional intervention before deal closure.

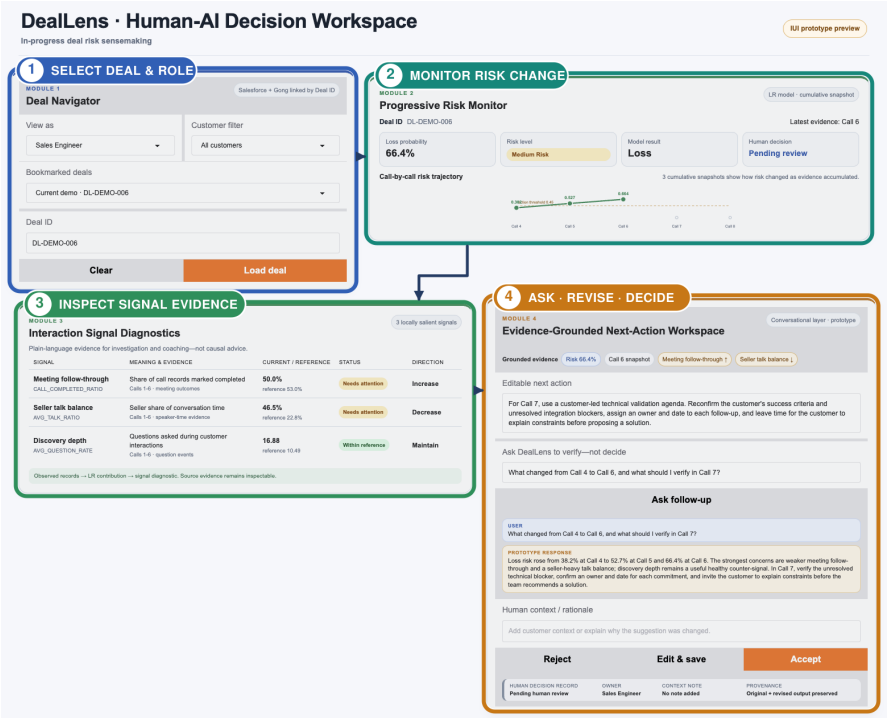


Figure 3: Interaction-layer workflow: users move through four connected modules retaining revision and decision authority.

4.2 Interaction Layer and Collaborative Workflow

The complete web interface implements four connected modules aligned with DG3 and DG4. (1) Deal Navigator supports Deal ID entry, customer filtering, and bookmarks to locate active deals. (2) Progressive Risk Monitor displays current Loss probability, risk level, and Call-4-onward trajectories so users can inspect change rather than an isolated score. (3) Interaction Signal Diagnostics ranks the three most influential current signals and pairs each with a plain-language definition, reference range, health state, and desired direction.(4)Evidence-Grounded Next-Action Workspace integrates an LLM API as its reasoning engine within a conversational UI, grounding AI guidance in selected risk and interaction evidence. Users can ask follow-up questions, accept, edit, or reject suggestions, add context, and review original and revised outputs. The interface is implemented, while its effects on decision quality and appropriate reliance remain to be evaluated.

4.3 User Story

The same evidence chain supports decisions at two user roles.

Sales Engineer prepares for the next call

After Call 5, an SE notices that the deal's loss risk has risen since Call 4. Rather than act on the score alone, the SE inspects the unusual interaction signals and asks DealLens what to verify in the next customer call. The SE rejects one suggestion using known customer context, edits the remaining action, and saves the reasoning for later review.

SE manager allocates limited support

During a weekly pipeline review, an SE manager compares two active opportunities: one shows worsening risk, while a large deal remains stable. After reviewing what changed and combining model evidence with team context, the manager assigns specialist support to the high-risk deal and maintains staffing for the stable one.

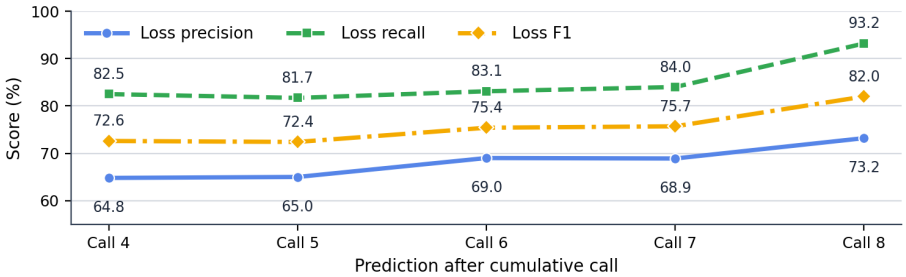


Figure 4: Loss-class precision, recall, and F1 across cumulative Call 4–8 snapshots; recall improves as evidence accumulates, motivating progressive rather than one-shot risk monitoring.

5 Results and Evaluation Roadmap

5.1 Predictive Readiness: Early and Progressive Loss Detection

DealLens demonstrates the feasibility of detecting deal risks early and refining predictions as conversational evidence accumulates. At Call 4, the model already identifies 82.5% of eventual losses (recall = 0.825; F1 = 0.726), providing actionable warning signals while opportunities remain active. As additional customer interactions unfold, Loss recall reaches 93.2% by Call 8, with F1 improving to 0.820. These results highlight the predictive value of cumulative interaction signals and the potential of longitudinal monitoring to reveal evolving deal risks before closure. Rather than automating sales decisions, DealLens transforms early risk detection into opportunities for evidence inspection, targeted coaching, and informed resource prioritization. Importantly, these findings establish predictive readiness, not causal business impact; calibration, false-alert costs, and improvements in actual sales outcomes require prospective field evaluation.

5.2 Planned User Study

No summative user study has been conducted. After interface completion and appropriate ethics review, we plan to recruit 8-12 participants with B2B Sales Engineering or closely related technical pre-sales experience, including individual contributors and managers. Using a progressive feature-ablation design, two groups will evaluate three systems each: Group 1 uses A -> C -> D to isolate the value of risk prediction and then dialogue; Group 2 uses B -> C -> D to isolate interaction diagnostics and then dialogue.

The study addresses three questions: whether risk trajectories improve deal prioritization (A vs. C), whether signal diagnostics improve evidence-based coaching (B vs. C), and whether conversational refinement and revision history support calibrated reliance beyond the static full system (C vs. D). We will examine triage choices, evidence cited, proposed actions, accepted/edited/rejected suggestions, decision time, workload, and participants' explanations of when they followed or overrode AI support.

Table 2: Planned progressive feature-ablation conditions comparing systems A–D to isolate the contributions of risk monitoring, signal diagnostics, and conversational refinement.

System	Available support	Comparison enabled
A	No Risk Monitor; Signal Diagnostics + static next-action suggestion	A vs. C: contribution of risk prediction
B	No Signal Diagnostics; Risk Monitor + static next-action suggestion	B vs. C: contribution of signal diagnostics
C	Risk Monitor + Signal Diagnostics + static next-action suggestion	C vs. D: contribution of conversational refinement
D	Full DealLens, including follow-up dialogue, comments, and revision history	Integrated human-AI workflow

Sessions combine timed tasks, think-aloud reasoning, and retrospective interviews. We will synchronize unaided and final judgments with evidence inspection, follow-up questions, confidence shifts, edits, override rationales, decision time, and NASA-TLX [23]. Analysis will trace judgment transitions before and after AI exposure, identifying justified reliance, missed counterevidence, and perceived agency across experience levels. Critical-incident coding will explain when professionals verify or contest advice; exploratory contrasts will report effect sizes and uncertainty, not definitive efficacy claims.

6 Limitation

DealLens integrates one organization's data, and only 195 of 1,868 labeled deals have four or more calls. Later-call performance is evaluated on smaller surviving cohorts: 93.2% Call-8 recall does not establish monotonic improvement for every deal. Time-ordered and cross-organization validation, per-stage sample sizes, and risk calibration are still needed.

The nine behavioral signals are observational associations, not causal levers; apparent risk shifts may reflect deal stage, account complexity, or customer availability. Call metrics must remain diagnostic evidence rather than individual performance ratings. Real-world deployment will require access controls, correction routes, and auditing to prevent decontextualized appraisal or surveillance [20,21].

Although all four interface modules are implemented, their effects on professional decisions are untested. The formative basis comprises two requirements sessions, not a saturated qualitative study. The planned 8–12-person evaluation can reveal interaction mechanisms but cannot establish generalizable efficacy. Long-term adoption, resource-allocation effects, and impacts on win rate or cycle time require larger field studies and prospective or defensible quasi-experimental evidence.

7 Conclusion and Future Work

DealLens advances enterprise decision support beyond retrospective analytics and isolated risk predictions toward continuous, evidence-grounded human-AI collaboration. By integrating CRM outcomes with conversational evidence, it demonstrates that emerging deal risks can be detected from early customer interactions and tracked as new evidence accumulates. Its implemented interface connects risk trajectories, interaction diagnostics, and conversational guidance within a unified workflow where professionals can inspect evidence, challenge interpretations, and revise suggested actions. Rather than replacing expertise with automated recommendations, DealLens establishes a technical and interaction foundation for making AI assistance traceable, contestable, and responsive to evolving professional context.

Future work will investigate how repeated AI assistance shapes professional judgment, authority, and capability over time. Controlled studies will examine when practitioners accept, question, or override AI advice, how different AI roles influence verification and decision ownership, and whether sustained assistance strengthens independent expertise or encourages overreliance. Longitudinal evaluations will further explore adaptive scaffolding—when AI should provide direct guidance, encourage reflection, or deliberately step back as professional capability develops. The goal is to move beyond immediate decision accuracy toward AI systems that improve judgment while preserving human agency and long-term independence.

A complementary direction will extend DealLens from call-level risk monitoring toward accountable agentic support for professional work. Future systems will identify pivotal conversational moments that shift risk assessments, elicit and reuse tacit expertise through human-approved memory, and audit differences between AI and professional reasoning. Building on these capabilities, evidence-to-artifact agents could transform validated judgments into contextualized meeting briefs, coaching plans, and executive reports through retrieval, clarification, provenance verification, and human revision. Together, these directions position DealLens as a foundation for investigating how intelligent systems can progress from predicting outcomes to supporting the development, application, and accountability of professional expertise.

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